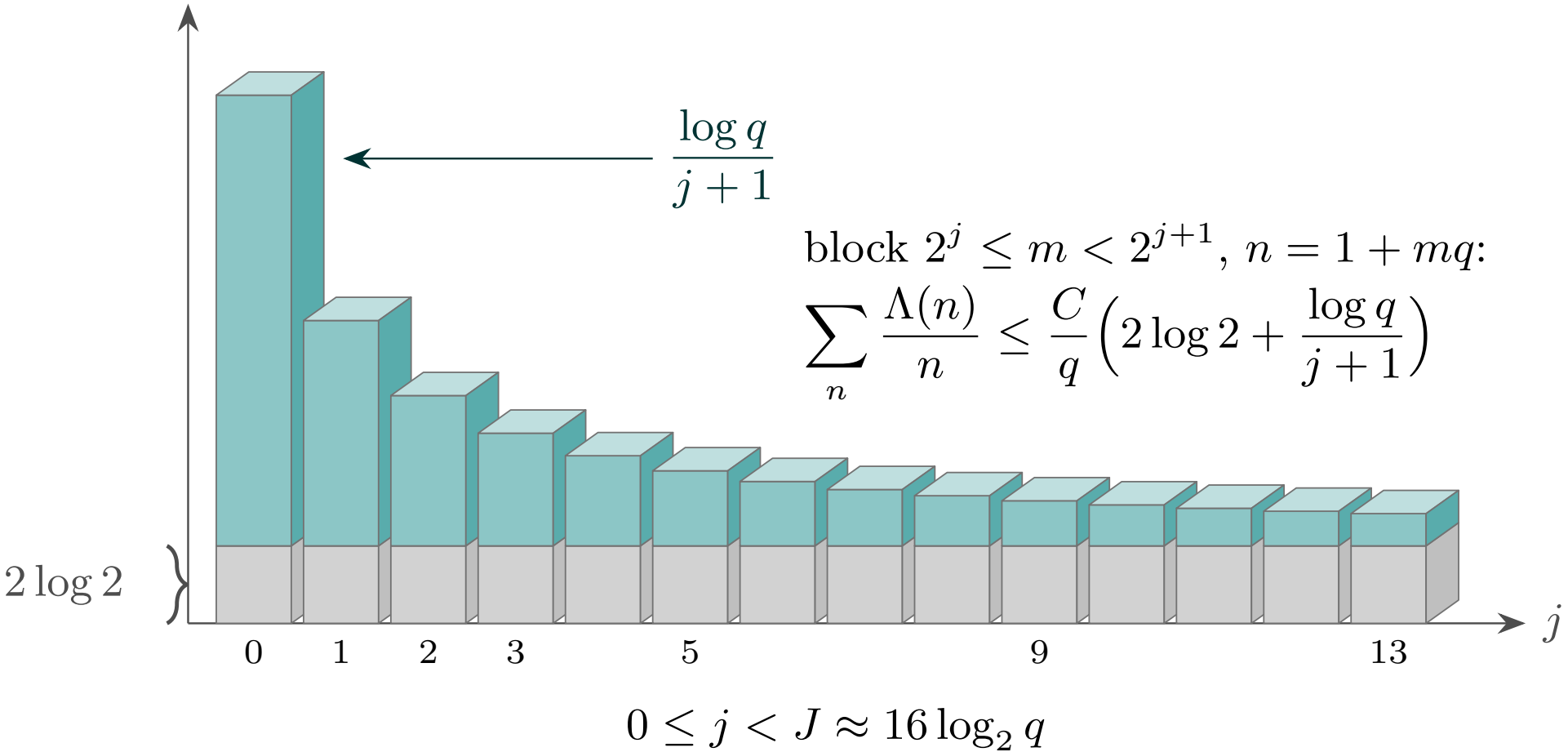

$\frac{\log q}{j+1}$
block $2^j \le m < 2^{j+1}$, $n = 1 + mq$:
$\sum_n \frac{\Lambda(n)}{n} \le \frac{C}{q}\left(2\log 2 + \frac{\log q}{j+1}\right)$
$2\log 2$
0
1
2
3
5
9
13
j
$0 \le j < J \approx 16 \log_2 q$